

Chapter Contents

Chapter 30 - Gamma And Bessel Functions	4
Why A New Function Appears	5
The Goal Of This Block	5
Bessel's Differential Equation	5
Classify The Origin	6
Where The Equation Comes From	6
Why Gamma Will Be Needed	7
Block 1 Summary	8
The Gamma Function	8
The Goal Of This Block	8
The Defining Integral	8
Derive The Gamma Recurrence	10
Recover Ordinary Factorials	11
The Half-Integer Starting Value	11
Move To Negative Nonintegers	12
Worked Gamma Calculation	13
Convert A Product Into Gamma Notation	13
Block 2 Summary	14
Reindexing Infinite Series Safely	14
The Goal Of This Block	14
A Dummy Index Can Be Renamed	14
Reindex A Shifted Power	15
Expose The Missing Low-Index Terms	15
Common Mistake: Changing The Exponent But Not The Coefficient	16
Block 3 Summary	17
Deriving The Bessel Function Of The First Kind	17
The Goal Of This Block	17
Begin With A Frobenius Series	17
Substitute Every Term	18
Reindex The Shifted Series	19
Find The Indicial Roots	20
Show Why The Odd Chain Vanishes	20
Derive The Later-Coefficient Recurrence	21
Build The Even Coefficients	22

Choose The Standard Normalization	23
Verify The Series Recurrence Locally	24
Block 4 Summary	25
Low Orders And The Second Solution	25
The Goal Of This Block	25
Construct J Zero	25
Construct J One	26
Construct J Two	26
Check J Zero In Its Equation	27
Use The Other Indicial Root	28
Why Integer Orders Need Y	29
The Order-Zero Second Solution	30
Block 5 Summary	30
Derivative And Neighbouring-Order Identities	31
The Goal Of This Block	31
Differentiate A Scaled Bessel Function	31
Extract The First Derivative Formula	32
The Companion Product Identity	33
Derive The Neighbour Sum	34
Derive The Neighbour Difference	35
Worked Identity Check At Order One	35
Block 6 Summary	36
Solving Equations By Transforming To Bessel Form	37
The Goal Of This Block	37
A Scaled Bessel Equation	37
Worked Scaled Equation	38
Apply A Boundedness Condition	39
The Spherical Bessel Equation	40
Complete The Spherical Transformation	40
Half-Integer Orders Become Trigonometric	42
Block 7 Summary	43
Behaviour, Zeros, And Boundary Conditions	43
The Goal Of This Block	43
Behaviour Near The Origin	43
Oscillation For Large X	44
Zeros Select Admissible Modes	44

Derivative Conditions Select Derivative Zeros	45
A Preview Of Orthogonality	46
Block 8 Summary	47
A Reliable Gamma And Bessel Workflow	47
The Goal Of This Block	47
The Workflow	47
Common Mistake: Confusing Gamma With Factorial	48
Common Mistake: Losing The New Lower Limit	48
Common Mistake: Using J Plus And J Minus At Integer Order	48
Common Mistake: Dropping A Chain-Rule Factor	49
Common Mistake: Rejecting A Singular Solution Too Early	49
Common Mistake: Treating An Asymptotic Formula As Equality	50
Block 9 Summary	50
Original Practice Set	50
Problems 1 To 4: Gamma And Series Algebra	50
Problems 5 To 8: Bessel Series And Identities	51
Problems 9 To 12: Differential Equations And Conditions	51
Worked Answers: Problems 1 To 4	52
Worked Answers: Problems 5 To 8	54
Worked Answers: Problems 9 To 12	56
Chapter 30 Final Summary	60

Chapter 30 - Gamma And Bessel Functions

Power-series solutions often contain products such as:

$$(\nu + 1)(\nu + 2) \cdots (\nu + m).$$

When ν is not an integer, ordinary factorial notation cannot describe this product cleanly. The gamma function supplies the missing notation. It then allows the Frobenius solution of Bessel's equation to be written as a single named function: the Bessel function J_ν .

This chapter will:

- define the gamma function and derive its recurrence
- connect gamma values with factorials and half-integer factorials
- make changes of index in infinite series explicit
- derive $J_\nu(x)$ from Bessel's differential equation by Frobenius' method
- construct the low-order functions J_0 , J_1 , and J_2
- explain when a second solution Y_ν is required
- derive the main derivative and neighbouring-order identities
- transform related differential equations into Bessel form
- interpret the behaviour and zeros of Bessel functions
- build a reliable workflow for solving Bessel-type equations

Why A New Function Appears

The Goal Of This Block

The goal is to recognise Bessel's equation and understand why its solutions cannot usually be written using only elementary functions.

a named special function is not a shortcut around the differential equation; it is a carefully normalized solution produced by the equation.

Bessel's Differential Equation

For a fixed real number ν , Bessel's equation of order ν is:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

The symbols have different jobs:

- x is the independent variable
- $y(x)$ is the unknown function
- ν is the order of the equation
- ν is fixed while the equation is being solved

Dividing by x^2 , for $x \neq 0$, gives normalized form:

$$y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right)y = 0.$$

The coefficients $1/x$ and ν^2/x^2 are singular at $x = 0$. This is why an ordinary Taylor series is not the natural first choice there.

Classify The Origin

Write the normalized equation as:

$$y'' + P(x)y' + Q(x)y = 0,$$

where:

$$P(x) = \frac{1}{x}, \quad Q(x) = 1 - \frac{\nu^2}{x^2}.$$

To test $x = 0$, calculate:

$$xP(x) = 1,$$

and:

$$x^2Q(x) = x^2 - \nu^2.$$

Both expressions are analytic at 0. Therefore:

$x = 0 \text{ is a regular singular point.}$

The Frobenius method from Chapter 28 is therefore the appropriate tool.

Where The Equation Comes From

Bessel's equation commonly appears when a problem has circular or cylindrical geometry. After variables are separated, the radial part often has the form:

$$r^2R'' + rR' + (\lambda^2r^2 - \nu^2)R = 0.$$

To convert this radial equation to standard Bessel form, set:

$$x = \lambda r.$$

The chain rule gives:

$$\frac{dR}{dr} = \lambda \frac{dR}{dx}, \quad \frac{d^2R}{dr^2} = \lambda^2 \frac{d^2R}{dx^2}.$$

Since $r = x/\lambda$, substitution produces:

$$x^2 R_{xx} + x R_x + (x^2 - \nu^2) R = 0.$$

Thus the same normalized equation describes many different radial models.

Why Gamma Will Be Needed

The Frobenius recurrence for Bessel's equation will generate products of the form:

$$(\nu + 1)(\nu + 2) \cdots (\nu + m).$$

If $\nu = 3$, this product can be written using factorials:

$$4 \cdot 5 \cdots (3 + m) = \frac{(m + 3)!}{3!}.$$

If $\nu = 1/2$, expressions such as $(m + 1/2)!$ are not ordinary factorials. The gamma function extends factorial notation to such values.

The dependency is:

gamma function $\rightarrow$ compact Frobenius coefficients $\rightarrow$ Bessel function.

Block 1 Summary

Bessel's equation has a regular singular point at the origin. Frobenius' method is therefore natural, and the resulting products require a factorial-like function that also accepts noninteger inputs.

The Gamma Function

The Goal Of This Block

The goal is to define $\Gamma(s)$, derive its recurrence without skipping the integration-by-parts boundary term, and use it to evaluate integer, half-integer, and negative noninteger arguments.

The Defining Integral

For $s > 0$, define:

$$\Gamma(s) = \int_0^{\infty} t^{s-1} e^{-t} dt.$$

The integration variable t is a dummy variable. It has no connection with the argument s .

Two competing factors make the integral finite:

- near $t = 0$, the power t^{s-1} is integrable when $s > 0$
- as $t \rightarrow \infty$, the exponential e^{-t} decays faster than any power grows

For example:

$$\Gamma(1) = \int_0^{\infty} e^{-t} dt.$$

Evaluate the improper integral:

$$\begin{aligned}\Gamma(1) &= \lim_{b \rightarrow \infty} \int_0^b e^{-t} dt \\ &= \lim_{b \rightarrow \infty} [-e^{-t}]_0^b \\ &= \lim_{b \rightarrow \infty} (1 - e^{-b}) \\ &= 1.\end{aligned}$$

Derive The Gamma Recurrence

Start from the definition with argument $s + 1$:

$$\Gamma(s + 1) = \int_0^\infty t^s e^{-t} dt.$$

Use integration by parts. Choose:

$$u = t^s, \quad dv = e^{-t} dt.$$

Then:

$$du = s t^{s-1} dt, \quad v = -e^{-t}.$$

Apply $\int u dv = uv - \int v du$ on $[0, b]$:

$$\int_0^b t^s e^{-t} dt = [-t^s e^{-t}]_0^b + s \int_0^b t^{s-1} e^{-t} dt.$$

Now take $b \rightarrow \infty$. For $s > 0$, both endpoint contributions vanish:

$$\lim_{b \rightarrow \infty} b^s e^{-b} = 0, \quad \lim_{t \rightarrow 0^+} t^s e^{-t} = 0.$$

Therefore the boundary term is zero, leaving:

$$\Gamma(s + 1) = s \int_0^\infty t^{s-1} e^{-t} dt.$$

Recognize the remaining integral as $\Gamma(s)$:

$$\boxed{\Gamma(s + 1) = s\Gamma(s)}.$$

The exponential wins over the power at infinity.¹

¹ Repeated use of l'Hopital's rule on b^s/e^b when s is an integer, or a standard exponential-growth estimate for general $s > 0$, gives $b^s e^{-b} \rightarrow 0$.

Recover Ordinary Factorials

Use the recurrence repeatedly:

$$\begin{aligned}
 \Gamma(n+1) &= n\Gamma(n) \\
 &= n(n-1)\Gamma(n-1) \\
 &= n(n-1)(n-2)\Gamma(n-2) \\
 &= \dots \\
 &= n(n-1)\dots 2 \cdot 1 \Gamma(1).
 \end{aligned}$$

Since $\Gamma(1) = 1$:

$$\boxed{\Gamma(n+1) = n!} \quad (n = 0, 1, 2, \dots).$$

Notice the shift:

$$\Gamma(n) = (n-1)!,$$

not $n!$. For example:

$$\Gamma(5) = 4! = 24.$$

The Half-Integer Starting Value

A fundamental value is:

$$\boxed{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}.$$

It follows from the Gaussian integral.² Once this starting value is known, the recurrence produces every positive half-integer value.

² With $t = u^2$, the gamma integral becomes half of the standard Gaussian integral, giving $\Gamma(1/2) = \sqrt{\pi}$.

For example:

$$\begin{aligned}\Gamma\left(\frac{3}{2}\right) &= \frac{1}{2}\Gamma\left(\frac{1}{2}\right) \\ &= \frac{\sqrt{\pi}}{2},\end{aligned}$$

and:

$$\begin{aligned}\Gamma\left(\frac{5}{2}\right) &= \frac{3}{2}\Gamma\left(\frac{3}{2}\right) \\ &= \frac{3}{2} \cdot \frac{\sqrt{\pi}}{2} \\ &= \frac{3\sqrt{\pi}}{4}.\end{aligned}$$

Move To Negative Nonintegers

Rearrange the recurrence:

$$\Gamma(s) = \frac{\Gamma(s+1)}{s}, \quad s \neq 0.$$

This relation extends gamma to negative noninteger arguments. For example, set $s = -1/2$:

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma(1/2)}{-1/2}.$$

Dividing by $-1/2$ is multiplying by -2 :

$$\boxed{\Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}}.$$

At $s = 0, -1, -2, \dots$, repeated use of the recurrence would require division by zero. Gamma has poles at these values; it is not finite there.

Worked Gamma Calculation

Evaluate $\Gamma(7/2)$ exactly.

Move downward one step at a time:

$$\Gamma\left(\frac{7}{2}\right) = \frac{5}{2}\Gamma\left(\frac{5}{2}\right).$$

Use the recurrence again:

$$\Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\Gamma\left(\frac{3}{2}\right).$$

And once more:

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right).$$

Substitute all three steps:

$$\begin{aligned}\Gamma\left(\frac{7}{2}\right) &= \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \\ &= \frac{5 \cdot 3 \cdot 1}{2^3} \sqrt{\pi} \\ &= \boxed{\frac{15\sqrt{\pi}}{8}}.\end{aligned}$$

Convert A Product Into Gamma Notation

For any ν for which the gamma values exist:

$$\Gamma(\nu + m + 1) = (\nu + m)\Gamma(\nu + m).$$

Continue the recurrence down to $\Gamma(\nu + 1)$:

$$\Gamma(\nu + m + 1) = (\nu + m)(\nu + m - 1) \cdots (\nu + 1)\Gamma(\nu + 1).$$

Divide by $\Gamma(\nu + 1)$:

$$(\nu + 1)(\nu + 2) \cdots (\nu + m) = \frac{\Gamma(\nu + m + 1)}{\Gamma(\nu + 1)}.$$

This is the exact product that will occur in the Bessel coefficients.

Block 2 Summary

The gamma function satisfies $\Gamma(s + 1) = s\Gamma(s)$, agrees with shifted factorials at integers, and makes noninteger factorial-like products precise.

Reindexing Infinite Series Safely

The Goal Of This Block

The goal is to make every change of index explicit before applying Frobenius' method to Bessel's equation.

changing an index changes the label and usually the lower limit; it does not change the terms represented by the series.

A Dummy Index Can Be Renamed

The letters k , m , and j are labels inside a summation. Therefore:

$$\sum_{k=0}^{\infty} a_k x^k = \sum_{j=0}^{\infty} a_j x^j.$$

Only the label changed. The lower limit, upper limit, coefficient subscript, and exponent still describe the same sequence of terms.

This is different from replacing k by $j - 2$, which is a genuine change of index and changes the limits.

Reindex A Shifted Power

Consider:

$$\sum_{k=0}^{\infty} a_k x^{k+2}.$$

We want the exponent to be j . Set:

$$j = k + 2.$$

Solve for the old index:

$$k = j - 2.$$

When $k = 0$, the new index is $j = 2$. Therefore:

$$\sum_{k=0}^{\infty} a_k x^{k+2} = \sum_{j=2}^{\infty} a_{j-2} x^j.$$

Finally rename j as k if desired:

$$\boxed{\sum_{k=0}^{\infty} a_k x^{k+2} = \sum_{k=2}^{\infty} a_{k-2} x^k}.$$

Expose The Missing Low-Index Terms

Suppose we need to combine:

$$\sum_{k=0}^{\infty} A_k x^k + \sum_{k=2}^{\infty} B_k x^k.$$

The second series has no x^0 or x^1 term. Separate those terms from the first series:

$$\sum_{k=0}^{\infty} A_k x^k = A_0 + A_1 x + \sum_{k=2}^{\infty} A_k x^k.$$

Now combine only the ranges that match:

$$A_0 + A_1 x + \sum_{k=2}^{\infty} (A_k + B_k) x^k.$$

This separation is what produces an indicial equation, an a_1 equation, and the later coefficient recurrence in the Bessel calculation.

Common Mistake: Changing The Exponent But Not The Coefficient

The incorrect step:

$$\sum_{k=0}^{\infty} a_k x^{k+2} \neq \sum_{k=2}^{\infty} a_k x^k$$

loses the relationship between each coefficient and its original term.

The correct substitution $j = k + 2$ changes both:

$$x^{k+2} \rightarrow x^j, \quad a_k \rightarrow a_{j-2}.$$

Hence:

$$\sum_{k=0}^{\infty} a_k x^{k+2} = \sum_{j=2}^{\infty} a_{j-2} x^j.$$

Block 3 Summary

To reindex safely, define the new index, solve for the old index, transform the coefficient subscript, and recalculate the lower limit before renaming the dummy variable.

Deriving The Bessel Function Of The First Kind

The Goal Of This Block

The goal is to derive the complete series for $J_\nu(x)$ from Bessel's equation, including the indicial equation, the odd-coefficient chain, the even coefficients, and the normalization.

Begin With A Frobenius Series

Start from Bessel's equation:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

Use:

$$y = \sum_{k=0}^{\infty} a_k x^{k+r}, \quad a_0 \neq 0.$$

Differentiate term by term:

$$y' = \sum_{k=0}^{\infty} (k+r) a_k x^{k+r-1},$$

The corresponding second derivative is:

$$y'' = \sum_{k=0}^{\infty} (k+r)(k+r-1)a_k x^{k+r-2}.$$

The order ν , Frobenius exponent r , and summation index k are three different quantities.

Substitute Every Term

Multiply the derivatives by their coefficients:

$$x^2 y'' = \sum_{k=0}^{\infty} (k+r)(k+r-1)a_k x^{k+r},$$

$$x y' = \sum_{k=0}^{\infty} (k+r)a_k x^{k+r},$$

$$-\nu^2 y = -\sum_{k=0}^{\infty} \nu^2 a_k x^{k+r},$$

and:

$$x^2 y = \sum_{k=0}^{\infty} a_k x^{k+r+2}.$$

Substitute these four expressions into the equation:

$$\begin{aligned} \sum_{k=0}^{\infty} [(k+r)(k+r-1) + (k+r) - \nu^2] a_k x^{k+r} \\ + \sum_{k=0}^{\infty} a_k x^{k+r+2} = 0. \end{aligned}$$

In the substituted expression, simplify the first bracket:

$$(k+r)(k+r-1) + (k+r) = (k+r)^2.$$

Therefore:

$$\sum_{k=0}^{\infty} [(k+r)^2 - \nu^2] a_k x^{k+r} + \sum_{k=0}^{\infty} a_k x^{k+r+2} = 0.$$

Reindex The Shifted Series

In the second sum, set:

$$j = k + 2.$$

Then $k = j - 2$, and the lower limit changes from $k = 0$ to $j = 2$:

$$\sum_{k=0}^{\infty} a_k x^{k+r+2} = \sum_{j=2}^{\infty} a_{j-2} x^{j+r}.$$

Rename j as k :

$$\sum_{k=0}^{\infty} a_k x^{k+r+2} = \sum_{k=2}^{\infty} a_{k-2} x^{k+r}.$$

Now separate the $k = 0$ and $k = 1$ terms from the first sum:

$$\begin{aligned} & (r^2 - \nu^2) a_0 x^r \\ & + [(r+1)^2 - \nu^2] a_1 x^{r+1} \\ & + \sum_{k=2}^{\infty} ([(k+r)^2 - \nu^2] a_k + a_{k-2}) x^{k+r} = 0. \end{aligned}$$

The three coefficient conditions are now visible.

Find The Indicial Roots

The lowest power is x^r . Its coefficient must vanish:

$$(r^2 - \nu^2)a_0 = 0.$$

Because $a_0 \neq 0$:

$$r^2 - \nu^2 = 0.$$

Factor:

$$(r - \nu)(r + \nu) = 0.$$

Therefore the indicial roots are:

$$\boxed{r_1 = \nu, \quad r_2 = -\nu.}$$

We first use the larger root $r = \nu$, assuming $\nu \geq 0$.

Show Why The Odd Chain Vanishes

The coefficient of x^{r+1} gives:

$$[(r+1)^2 - \nu^2]a_1 = 0.$$

Set $r = \nu$:

$$[(\nu+1)^2 - \nu^2]a_1 = 0.$$

Expand the bracket:

$$(\nu^2 + 2\nu + 1 - \nu^2)a_1 = 0.$$

Thus:

$$(2\nu + 1)a_1 = 0.$$

For $\nu \geq 0$, the factor $2\nu + 1$ is nonzero. Divide by it:

$$\boxed{a_1 = 0.}$$

The later recurrence moves in steps of two, so $a_3, a_5, \dots$ will also be zero.

Derive The Later-Coefficient Recurrence

For $k \geq 2$:

$$[(k+r)^2 - \nu^2]a_k + a_{k-2} = 0.$$

Move a_{k-2} to the right:

$$[(k+r)^2 - \nu^2]a_k = -a_{k-2}.$$

Set $r = \nu$:

$$[(k+\nu)^2 - \nu^2]a_k = -a_{k-2}.$$

Factor the difference of squares:

$$(k + \nu)^2 - \nu^2 = k(k + 2\nu).$$

Divide by $k(k + 2\nu)$:

$$a_k = -\frac{a_{k-2}}{k(k + 2\nu)}, \quad k \geq 2.$$

Build The Even Coefficients

Set $k = 2$ in the recurrence:

$$\begin{aligned} a_2 &= -\frac{a_0}{2(2 + 2\nu)} \\ &= -\frac{a_0}{2^2(\nu + 1)}. \end{aligned}$$

Set $k = 4$:

$$\begin{aligned} a_4 &= -\frac{a_2}{4(4 + 2\nu)} \\ &= -\frac{a_2}{2^3(\nu + 2)} \\ &= \frac{a_0}{2^2(\nu + 1) 2^3(\nu + 2)} \\ &= \frac{a_0}{2^4(2!)(\nu + 1)(\nu + 2)}. \end{aligned}$$

The general even coefficient is:

$$a_{2m} = \frac{(-1)^m a_0}{2^{2m} m! (\nu + 1)(\nu + 2) \cdots (\nu + m)}.$$

Use the gamma product from Block 2:

$$(\nu + 1) \cdots (\nu + m) = \frac{\Gamma(\nu + m + 1)}{\Gamma(\nu + 1)}.$$

Therefore:

$$a_{2m} = \frac{(-1)^m a_0 \Gamma(\nu + 1)}{2^{2m} m! \Gamma(\nu + m + 1)}.$$

Choose The Standard Normalization

The Frobenius solution is:

$$y = x^\nu \sum_{m=0}^{\infty} a_{2m} x^{2m}.$$

Substitute the coefficient formula:

$$y = a_0 \Gamma(\nu + 1) x^\nu \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m} m! \Gamma(m + \nu + 1)}.$$

Choose:

$$a_0 = \frac{1}{2^\nu \Gamma(\nu + 1)}.$$

Then $a_0 \Gamma(\nu + 1) x^\nu = x^\nu / 2^\nu = (x/2)^\nu$. Combining this factor with $x^{2m} / 2^{2m}$ gives the standard Bessel function:

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m + \nu + 1)} \left(\frac{x}{2}\right)^{2m+\nu}.$$

The normalization is chosen so that, near $x = 0$:

$$J_\nu(x) \sim \frac{1}{\Gamma(\nu + 1)} \left(\frac{x}{2}\right)^\nu.$$

Verify The Series Recurrence Locally

Let the coefficient of $x^{2m+\nu}$ in J_ν be:

$$c_m = \frac{(-1)^m}{2^{2m+\nu} m! \Gamma(m + \nu + 1)}.$$

Compare c_m with c_{m-1} :

$$\frac{c_m}{c_{m-1}} = -\frac{2^{2m-2+\nu} (m-1)! \Gamma(m + \nu)}{2^{2m+\nu} m! \Gamma(m + \nu + 1)}.$$

Use:

$$m! = m(m-1)!, \quad \Gamma(m + \nu + 1) = (m + \nu) \Gamma(m + \nu).$$

After cancellation:

$$\frac{c_m}{c_{m-1}} = -\frac{1}{4m(m + \nu)}.$$

This is the even-index recurrence with $k = 2m$:

$$-\frac{1}{k(k + 2\nu)} = -\frac{1}{(2m)(2m + 2\nu)} = -\frac{1}{4m(m + \nu)}.$$

Thus the compact gamma formula preserves the recurrence derived from the differential equation.

Block 4 Summary

Frobenius' method gives indicial roots $\pm\nu$. The root $r = \nu$ produces an even coefficient chain whose gamma-normalized series is $J_\nu(x)$.

Low Orders And The Second Solution

The Goal Of This Block

The goal is to construct familiar low-order Bessel functions and explain why J_ν alone does not always provide the full two-dimensional solution space.

Construct J Zero

Set $\nu = 0$ in the defining series:

$$J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+1)} \left(\frac{x}{2}\right)^{2m}.$$

Since $\Gamma(m+1) = m!$:

$$J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2} \left(\frac{x}{2}\right)^{2m}.$$

Write the first four terms:

$$\begin{aligned} J_0(x) &= 1 - \frac{x^2}{2^2} + \frac{x^4}{2^4(2!)^2} - \frac{x^6}{2^6(3!)^2} + \cdots \\ &= \boxed{1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \cdots}. \end{aligned}$$

Construct J One

Set $\nu = 1$:

$$J_1(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+2)} \left(\frac{x}{2}\right)^{2m+1}.$$

Because $\Gamma(m+2) = (m+1)!$:

$$J_1(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!(m+1)!} \left(\frac{x}{2}\right)^{2m+1}.$$

Now calculate successive terms:

$$m = 0 : \quad \frac{x}{2},$$

$$m = 1 : \quad -\frac{1}{1!2!} \left(\frac{x}{2}\right)^3 = -\frac{x^3}{16},$$

$$m = 2 : \quad \frac{1}{2!3!} \left(\frac{x}{2}\right)^5 = \frac{x^5}{384}.$$

Therefore:

$$J_1(x) = \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} - \dots$$

Construct J Two

Set $\nu = 2$:

$$J_2(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+3)} \left(\frac{x}{2}\right)^{2m+2}.$$

Use $\Gamma(m+3) = (m+2)!$:

$$J_2(x) = \frac{1}{2!} \left(\frac{x}{2}\right)^2 - \frac{1}{1!3!} \left(\frac{x}{2}\right)^4 + \frac{1}{2!4!} \left(\frac{x}{2}\right)^6 - \dots$$

$$= \boxed{\frac{x^2}{8} - \frac{x^4}{96} + \frac{x^6}{3072} - \dots}.$$

Near the origin, J_2 behaves like $x^2/8$. More generally, for $\nu > 0$, $J_\nu(x)$ tends to zero like x^ν .

Check J Zero In Its Equation

The order-zero equation is:

$$x^2 y'' + x y' + x^2 y = 0.$$

Use the truncated series:

$$y = 1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6).$$

Differentiate:

$$y' = -\frac{x}{2} + \frac{x^3}{16} + O(x^5),$$

$$y'' = -\frac{1}{2} + \frac{3x^2}{16} + O(x^4).$$

Substitute each contribution into $x^2y'' + xy' + x^2y$:

$$x^2y'' = -\frac{x^2}{2} + \frac{3x^4}{16} + O(x^6),$$

$$xy' = -\frac{x^2}{2} + \frac{x^4}{16} + O(x^6),$$

$$x^2y = x^2 - \frac{x^4}{4} + O(x^6).$$

Add like powers:

$$\begin{aligned} x^2y'' + xy' + x^2y &= \left(-\frac{1}{2} - \frac{1}{2} + 1\right)x^2 \\ &\quad + \left(\frac{3}{16} + \frac{1}{16} - \frac{1}{4}\right)x^4 + O(x^6) \\ &= O(x^6). \end{aligned}$$

The displayed coefficients cancel exactly. Continuing the full recurrence cancels every higher power.

Use The Other Indicial Root

The second indicial root is $r = -\nu$. Repeating the Frobenius construction formally gives $J_{-\nu}(x)$:

$$J_{-\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m-\nu+1)} \left(\frac{x}{2}\right)^{2m-\nu}.$$

If ν is not an integer, J_ν and $J_{-\nu}$ are generally linearly independent. On an interval with $x > 0$, the general solution may then be written:

$$y = C_1 J_\nu(x) + C_2 J_{-\nu}(x).$$

Their different leading powers, x^ν and $x^{-\nu}$, make the independence visible near the origin when $\nu > 0$.

Why Integer Orders Need Y

For an integer n :

$$J_{-n}(x) = (-1)^n J_n(x).$$

Thus J_n and J_{-n} point in the same solution direction. They cannot form a basis for a second-order equation.

A second independent solution is called the Bessel function of the second kind and is denoted $Y_\nu(x)$. For noninteger ν , it can be defined by:

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\pi\nu) - J_{-\nu}(x)}{\sin(\pi\nu)}.$$

Integer orders are obtained by taking a limit as $\nu \rightarrow n$. Therefore, on an interval not containing 0:

$$\boxed{y = C_1 J_n(x) + C_2 Y_n(x)}.$$

The function J_n remains finite at the origin, while Y_n is singular there. A physical condition requiring boundedness at the centre commonly forces $C_2 = 0$.

The Order-Zero Second Solution

For $\nu = 0$, the indicial root is repeated. Frobenius theory warns that the second solution may contain a logarithm. That is exactly what happens:

$$Y_0(x) = \frac{2}{\pi} \left(\ln \frac{x}{2} + \gamma \right) J_0(x) + \text{a regular power series},$$

where γ is the Euler–Mascheroni constant.

The important structural point is not the full coefficient formula. It is:

$$Y_0(x) \sim \frac{2}{\pi} \ln x \quad (x \rightarrow 0^+).$$

Therefore Y_0 is unbounded at the origin, whereas $J_0(0) = 1$.

Block 5 Summary

The first-kind solutions $J_0, J_1, J_2, \dots$ come directly from the gamma series. Noninteger orders can use J_ν and $J_{-\nu}$ as a basis, but integer orders require the independent second-kind solution Y_n .

Derivative And Neighbouring-Order Identities

The Goal Of This Block

The goal is to derive a small set of identities that replace repeated differentiation of the full infinite series.

the order behaves like an index in a polynomial family: differentiation connects neighbouring orders.

Differentiate A Scaled Bessel Function

Start with the series for J_ν :

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+\nu}}{2^{2m+\nu} m! \Gamma(m+\nu+1)}.$$

Multiply by $x^{-\nu}$:

$$x^{-\nu} J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m+\nu} m! \Gamma(m+\nu+1)}.$$

Differentiate term by term. The $m = 0$ term has derivative zero, so begin at $m = 1$:

$$\frac{d}{dx} (x^{-\nu} J_\nu(x)) = \sum_{m=1}^{\infty} \frac{(-1)^m (2m) x^{2m-1}}{2^{2m+\nu} m! \Gamma(m+\nu+1)}.$$

Use $m! = m(m-1)!$ and cancel m :

$$= \sum_{m=1}^{\infty} \frac{(-1)^m x^{2m-1}}{2^{2m+\nu-1} (m-1)! \Gamma(m+\nu+1)}.$$

Set $j = m - 1$. Then $m = j + 1$ and the lower limit becomes $j = 0$:

$$= - \sum_{j=0}^{\infty} \frac{(-1)^j x^{2j+1}}{2^{2j+\nu+1} j! \Gamma(j + \nu + 2)}.$$

The series on the right is $x^{-\nu} J_{\nu+1}(x)$. Therefore:

$$\boxed{\frac{d}{dx} (x^{-\nu} J_{\nu}(x)) = -x^{-\nu} J_{\nu+1}(x)}.$$

Extract The First Derivative Formula

Expand the derivative on the left using the product rule:

$$\frac{d}{dx} (x^{-\nu} J_{\nu}) = -\nu x^{-\nu-1} J_{\nu} + x^{-\nu} J'_{\nu}.$$

Set this equal to the identity just derived:

$$-\nu x^{-\nu-1} J_{\nu} + x^{-\nu} J'_{\nu} = -x^{-\nu} J_{\nu+1}.$$

Multiply every term by x^{ν} :

$$-\frac{\nu}{x} J_{\nu} + J'_{\nu} = -J_{\nu+1}.$$

Move the first term to the right:

$$\boxed{J'_{\nu}(x) = \frac{\nu}{x} J_{\nu}(x) - J_{\nu+1}(x)}.$$

The Companion Product Identity

To derive the companion identity, multiply the defining series by x^ν :

$$x^\nu J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+2\nu}}{2^{2m+\nu} m! \Gamma(m+\nu+1)}.$$

Differentiate term by term:

$$\frac{d}{dx} (x^\nu J_\nu(x)) = \sum_{m=0}^{\infty} \frac{(-1)^m (2m+2\nu) x^{2m+2\nu-1}}{2^{2m+\nu} m! \Gamma(m+\nu+1)}.$$

Factor $2m+2\nu = 2(m+\nu)$ and use:

$$\Gamma(m+\nu+1) = (m+\nu)\Gamma(m+\nu).$$

Cancel the common factor $m+\nu$:

$$\frac{d}{dx} (x^\nu J_\nu(x)) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+2\nu-1}}{2^{2m+\nu-1} m! \Gamma(m+\nu)}.$$

The right side is $x^\nu J_{\nu-1}(x)$. Therefore:

$$\boxed{\frac{d}{dx} (x^\nu J_\nu(x)) = x^\nu J_{\nu-1}(x)}.$$

Expand its left side:

$$\nu x^{\nu-1} J_\nu + x^\nu J'_\nu = x^\nu J_{\nu-1}.$$

From $\nu x^{\nu-1}J_\nu + x^\nu J'_\nu = x^\nu J_{\nu-1}$, divide every term by x^ν :

$$\frac{\nu}{x}J_\nu + J'_\nu = J_{\nu-1}.$$

Solve for the derivative:

$$J'_\nu(x) = J_{\nu-1}(x) - \frac{\nu}{x}J_\nu(x).$$

The two derivative formulas are equivalent descriptions of the same function.

Derive The Neighbour Sum

We now have:

$$J'_\nu = \frac{\nu}{x}J_\nu - J_{\nu+1},$$

The second derivative identity is:

$$J'_\nu = J_{\nu-1} - \frac{\nu}{x}J_\nu.$$

Set the right sides equal:

$$\frac{\nu}{x}J_\nu - J_{\nu+1} = J_{\nu-1} - \frac{\nu}{x}J_\nu.$$

Add $J_{\nu+1}$ and $\nu J_\nu/x$ to both sides:

$$\frac{2\nu}{x}J_\nu = J_{\nu-1} + J_{\nu+1}.$$

The rearrangement gives the neighbour sum:

$$J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_{\nu}(x).$$

Derive The Neighbour Difference

Use the same two derivative equations:

$$J'_{\nu} = \frac{\nu}{x} J_{\nu} - J_{\nu+1},$$

and:

$$J'_{\nu} = J_{\nu-1} - \frac{\nu}{x} J_{\nu}.$$

Add the equations. On the right, the two $\nu J_{\nu}/x$ terms cancel:

$$\begin{aligned} 2J'_{\nu} &= \frac{\nu}{x} J_{\nu} - J_{\nu+1} + J_{\nu-1} - \frac{\nu}{x} J_{\nu} \\ &= J_{\nu-1} - J_{\nu+1}. \end{aligned}$$

Divide by 2:

$$J'_{\nu}(x) = \frac{1}{2} [J_{\nu-1}(x) - J_{\nu+1}(x)].$$

Worked Identity Check At Order One

Set $\nu = 1$ in the neighbour sum:

$$J_0(x) + J_2(x) = \frac{2}{x} J_1(x).$$

Use the series through terms of degree four:

$$J_0(x) = 1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6),$$

$$J_2(x) = \frac{x^2}{8} - \frac{x^4}{96} + O(x^6).$$

Add them:

$$\begin{aligned} J_0 + J_2 &= 1 + \left(-\frac{1}{4} + \frac{1}{8}\right)x^2 + \left(\frac{1}{64} - \frac{1}{96}\right)x^4 + O(x^6) \\ &= 1 - \frac{x^2}{8} + \frac{x^4}{192} + O(x^6). \end{aligned}$$

Now use:

$$J_1(x) = \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} + O(x^7).$$

Multiply by $2/x$:

$$\begin{aligned} \frac{2}{x}J_1(x) &= \frac{2}{x} \left(\frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} \right) + O(x^6) \\ &= 1 - \frac{x^2}{8} + \frac{x^4}{192} + O(x^6). \end{aligned}$$

Both sides agree term by term.

Block 6 Summary

The product identities yield derivative and neighbouring-order recurrences. These formulas let us move among $J_{\nu-1}$, J_ν , $J_{\nu+1}$, and J'_ν without rebuilding an infinite series.

Solving Equations By Transforming To Bessel Form

The Goal Of This Block

The goal is to identify a scaled Bessel equation, carry the chain rule through the substitution, and handle spherical Bessel equations using a dependent- variable transformation.

A Scaled Bessel Equation

Consider:

$$x^2 y'' + xy' + (\lambda^2 x^2 - \nu^2)y = 0, \quad \lambda > 0.$$

Define a new independent variable:

$$z = \lambda x,$$

For $z = \lambda x$, write the dependent variable as:

$$y(x) = u(z).$$

The first derivative is:

$$\frac{dy}{dx} = \frac{du}{dz} \frac{dz}{dx} = \lambda u_z.$$

Differentiate again:

$$\frac{d^2 y}{dx^2} = \lambda \frac{d}{dx}(u_z) = \lambda u_{zz} \frac{dz}{dx} = \lambda^2 u_{zz}.$$

Also $x = z/\lambda$. These three substitutions must all be used together.

Worked Scaled Equation

Solve on $x > 0$:

$$x^2 y'' + xy' + (4x^2 - 9)y = 0.$$

Compare with the scaled form:

$$x^2 y'' + xy' + (\lambda^2 x^2 - \nu^2)y = 0.$$

Match coefficients:

$$\lambda^2 = 4, \quad \nu^2 = 9.$$

Choose $\lambda = 2$ and order $\nu = 3$. Set:

$$z = 2x, \quad y(x) = u(z).$$

Using $z = 2x$, the derivative and variable substitutions are:

$$y' = 2u_z, \quad y'' = 4u_{zz}, \quad x = \frac{z}{2}.$$

Substitute into the original equation:

$$\left(\frac{z}{2}\right)^2 (4u_{zz}) + \left(\frac{z}{2}\right) (2u_z) + \left[4\left(\frac{z}{2}\right)^2 - 9\right] u = 0.$$

Simplify each coefficient:

$$z^2 u_{zz} + zu_z + (z^2 - 9)u = 0.$$

The simplified equation is:

$$z^2 u_{zz} + zu_z + (z^2 - 9)u = 0.$$

This is Bessel's equation of order 3. Therefore:

$$u(z) = C_1 J_3(z) + C_2 Y_3(z).$$

Return to $z = 2x$:

$$y(x) = C_1 J_3(2x) + C_2 Y_3(2x).$$

Apply A Boundedness Condition

Suppose the previous equation is used for $x \geq 0$ and the solution must remain bounded at $x = 0$.

Near zero:

$$J_3(2x) \sim \frac{1}{\Gamma(4)} x^3 = \frac{x^3}{6},$$

so $J_3(2x)$ remains finite and tends to zero. In contrast, $Y_3(2x)$ is singular at zero.

Therefore boundedness forces:

$$C_2 = 0.$$

The admissible family becomes:

$$y(x) = C_1 J_3(2x).$$

The condition did not solve for C_1 because every multiple of a bounded solution is still bounded. A normalization or another boundary condition would be needed to determine it.

The Spherical Bessel Equation

The spherical Bessel equation is:

$$x^2 u'' + 2xu' + [x^2 - \ell(\ell + 1)]u = 0.$$

The coefficient of xu' is 2, not 1, so this is not yet ordinary Bessel form. Remove the mismatch by setting:

$$u(x) = x^{-1/2}v(x).$$

Differentiate using the product rule:

$$u' = -\frac{1}{2}x^{-3/2}v + x^{-1/2}v',$$

and differentiate once more:

$$u'' = \frac{3}{4}x^{-5/2}v - x^{-3/2}v' + x^{-1/2}v''.$$

The middle coefficient -1 comes from adding two copies of $-(1/2)x^{-3/2}v'$.

Complete The Spherical Transformation

Substitute the derivatives into:

$$x^2 u'' + 2xu' + [x^2 - \ell(\ell + 1)]u = 0.$$

First calculate:

$$x^2 u'' = \frac{3}{4}x^{-1/2}v - x^{1/2}v' + x^{3/2}v'',$$

The contribution from $2xu'$ is:

$$2xu' = -x^{-1/2}v + 2x^{1/2}v'.$$

The final term is:

$$[x^2 - \ell(\ell + 1)]u = [x^2 - \ell(\ell + 1)]x^{-1/2}v.$$

Add the three expressions and factor out $x^{-1/2}$:

$$x^{-1/2} \left\{ x^2 v'' + xv' + \left[x^2 - \ell(\ell + 1) - \frac{1}{4} \right] v \right\} = 0.$$

For $x > 0$, multiply by $x^{1/2}$:

$$x^2 v'' + xv' + \left[x^2 - \ell(\ell + 1) - \frac{1}{4} \right] v = 0.$$

Complete the square in the order term:

$$\ell(\ell + 1) + \frac{1}{4} = \ell^2 + \ell + \frac{1}{4} = \left(\ell + \frac{1}{2} \right)^2.$$

Therefore:

$$x^2 v'' + xv' + \left[x^2 - \left(\ell + \frac{1}{2} \right)^2 \right] v = 0.$$

This is Bessel's equation of order $\ell + 1/2$.

Half-Integer Orders Become Trigonometric

The first half-integer functions are:

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x,$$

The second half-integer function is:

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x.$$

For $\ell = 0$, the spherical equation is:

$$x^2 u'' + 2xu' + x^2 u = 0.$$

Since $u = x^{-1/2}v$ and v has order $1/2$:

$$u(x) = x^{-1/2} [C_1 J_{1/2}(x) + C_2 J_{-1/2}(x)].$$

Insert the trigonometric forms:

$$u(x) = \sqrt{\frac{2}{\pi}} \left[C_1 \frac{\sin x}{x} + C_2 \frac{\cos x}{x} \right].$$

Absorb the common nonzero factor $\sqrt{2/\pi}$ into the arbitrary constants:

$$\boxed{u(x) = A \frac{\sin x}{x} + B \frac{\cos x}{x}}.$$

Block 7 Summary

A scaled independent variable handles $\lambda^2 x^2$, while a power of x can correct the first-derivative coefficient. Every derivative and return substitution must be shown explicitly.

Behaviour, Zeros, And Boundary Conditions

The Goal Of This Block

The goal is to connect the formulas with the shape of Bessel solutions and with the boundary conditions that select particular orders or zeros.

Behaviour Near The Origin

The first term of the J_ν series is:

$$J_\nu(x) = \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu + \text{higher powers.}$$

Therefore, for $\nu \geq 0$:

$$J_0(0) = 1,$$

while:

$$J_\nu(0) = 0 \quad (\nu > 0).$$

The second-kind solution behaves differently:

- $Y_0(x)$ has a logarithmic singularity
- $Y_\nu(x)$ behaves like a negative power for $\nu > 0$

This local behaviour often decides which solution survives a condition at the centre before any constants are calculated.

Oscillation For Large x

For fixed ν and large positive x :

$$J_\nu(x) \sim \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right).$$

This formula has two visible parts:

- the cosine shows oscillation
- the factor $x^{-1/2}$ shows a slowly decreasing amplitude

Unlike $\sin x$, a Bessel wave does not oscillate with constant amplitude. The geometry encoded by the xy' term spreads the oscillation as x grows.

Zeros Select Admissible Modes

Suppose a bounded radial solution on $0 \leq r \leq R$ has the form:

$$u(r) = AJ_n(\lambda r).$$

Impose the outer boundary condition:

$$u(R) = 0.$$

Substitute $r = R$:

$$AJ_n(\lambda R) = 0.$$

For a nontrivial solution, $A \neq 0$. Divide by A :

$$J_n(\lambda R) = 0.$$

Let $j_{n,k}$ denote the k th positive zero of J_n . Then:

$$\lambda R = j_{n,k}.$$

Solve for λ :

$$\lambda_k = \frac{j_{n,k}}{R}.$$

The boundary does not allow every value of λ ; it selects a discrete set of modes.

Derivative Conditions Select Derivative Zeros

If the outer condition is instead:

$$u'(R) = 0,$$

differentiate $u(r) = AJ_n(\lambda r)$ using the chain rule:

$$u'(r) = A\lambda J'_n(\lambda r).$$

Set $r = R$:

$$A\lambda J'_n(\lambda R) = 0.$$

For a nontrivial mode with $A \neq 0$ and $\lambda \neq 0$:

$$J'_n(\lambda R) = 0.$$

Thus value conditions involve zeros of J_n , while derivative conditions involve zeros of J'_n .

A Preview Of Orthogonality

Write the scaled Bessel equation as:

$$r^2 u'' + r u' + (\lambda^2 r^2 - n^2) u = 0.$$

Divide by r :

$$r u'' + u' + \left(\lambda^2 r - \frac{n^2}{r} \right) u = 0.$$

Recognize the product derivative:

$$(r u')' = r u'' + u'.$$

Hence:

$$(r u')' + \left(\lambda^2 r - \frac{n^2}{r} \right) u = 0.$$

This is self-adjoint form with weight $w(r) = r$. Under suitable boundary conditions, modes corresponding to different values of λ are orthogonal with respect to:

$$\langle f, g \rangle = \int_0^R f(r) g(r) r \, dr.$$

The factor r is geometric: a thin circular ring has area proportional to its radius.

Block 8 Summary

First-kind solutions are regular at the centre, large- x solutions oscillate with decaying amplitude, and outer boundary conditions select zeros of either J_n or J'_n .

A Reliable Gamma And Bessel Workflow

The Goal Of This Block

The goal is to turn the chapter into a repeatable decision process and identify the errors most likely to break a derivation.

The Workflow

Step 1: Identify the task. Decide whether the problem concerns a gamma value, a Bessel series, an identity, or a differential equation.

Step 2: For gamma, move toward a known argument. Use:

$$\Gamma(s+1) = s\Gamma(s),$$

until the argument reaches 1, a positive integer, or $1/2$.

Step 3: For a differential equation, compare all three coefficients. Look for:

$$x^2y'' + xy' + (\lambda^2x^2 - \nu^2)y = 0.$$

Step 4: Show the variable change. If $z = \lambda x$, calculate both y' and y'' before substituting.

Step 5: Choose the correct basis. Use J_ν, Y_ν for integer order on an interval away from zero, or $J_\nu, J_{-\nu}$ for noninteger order when useful.

Step 6: Apply centre behaviour before outer conditions. Boundedness at $x = 0$ usually removes the Y_ν term.

Step 7: Use identities locally. State the derivative or neighbour formula immediately before substituting its order.

Step 8: Return to the original variable. A solution in z is incomplete until $z = \lambda x$ has been restored.

Common Mistake: Confusing Gamma With Factorial

The relationship is:

$$\Gamma(n + 1) = n!.$$

Therefore:

$$\Gamma(6) = 5! = 120,$$

not $6! = 720$. Always reduce the gamma argument by one before replacing it with a factorial.

Common Mistake: Losing The New Lower Limit

Under $j = k + 2$:

$$k = 0 \quad \rightarrow \quad j = 2.$$

Therefore:

$$\sum_{k=0}^{\infty} a_k x^{k+r+2} = \sum_{j=2}^{\infty} a_{j-2} x^{j+r}.$$

Starting the new sum at 0 invents nonexistent coefficients a_{-2} and a_{-1} and hides the indicial and a_1 equations.

Common Mistake: Using J Plus And J Minus At Integer Order

For integer n :

$$J_{-n} = (-1)^n J_n.$$

Using $J_{-n} = (-1)^n J_n$, the proposed combination becomes:

$$C_1 J_n + C_2 J_{-n} = [C_1 + (-1)^n C_2] J_n,$$

which contains only one independent direction. Use J_n and Y_n for the general solution instead.

Common Mistake: Dropping A Chain-Rule Factor

If $z = 3x$ and $y(x) = u(z)$:

$$y' = 3u_z,$$

but the second derivative is:

$$y'' = 9u_{zz},$$

not $3u_{zz}$. The factor appears once for each differentiation with respect to x .

Common Mistake: Rejecting A Singular Solution Too Early

Y_ν is singular at the origin, but that does not make it invalid on every domain. On an annulus or any interval $a < x < b$ with $a > 0$, the origin is outside the domain and both J_ν and Y_ν may be needed.

Reject a solution because it violates the stated domain or condition, not because it is singular somewhere the problem never includes.

Common Mistake: Treating An Asymptotic Formula As Equality

The notation:

$$J_\nu(x) \sim \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\nu\pi}{2} - \frac{\pi}{4}\right)$$

describes leading behaviour as $x \rightarrow \infty$. It is not an exact identity for finite x . Use the defining series, recurrence identities, or numerical evaluation when exact finite- x information is required.

Block 9 Summary

Reliable work depends on preserving index limits, gamma shifts, chain-rule factors, solution independence, domains, and the distinction between exact and asymptotic formulas.

Original Practice Set

Problems 1 To 4: Gamma And Series Algebra

Problem 1: Evaluate Gamma At An Integer. Find $\Gamma(8)$.

Problem 2: Evaluate A Half-Integer. Find $\Gamma(9/2)$ exactly.

Problem 3: Move To A Negative Argument. Find $\Gamma(-3/2)$ exactly.

Problem 4: Reindex A Series. Rewrite:

$$\sum_{k=0}^{\infty} (k+1)a_k x^{k+3}$$

as a series in powers x^j beginning at the correct lower limit.

Problems 5 To 8: Bessel Series And Identities

Problem 5: Build A Series. Write the first four nonzero terms of $J_3(x)$.

Problem 6: Check The Leading Behaviour. Use the defining series to show:

$$J_{5/2}(x) \sim \frac{1}{\Gamma(7/2)} \left(\frac{x}{2}\right)^{5/2} \quad (x \rightarrow 0^+).$$

Then simplify the coefficient using $\Gamma(7/2)$.

Problem 7: Use A Neighbour Recurrence. Express $J_3(x)$ in terms of $J_2(x)$ and $J_1(x)$.

Problem 8: Differentiate Without The Full Series. Express $J_2'(x)$ in terms of $J_1(x)$ and $J_3(x)$.

Problems 9 To 12: Differential Equations And Conditions

Problem 9: Solve A Scaled Equation. Solve on $x > 0$:

$$x^2 y'' + xy' + (25x^2 - 4)y = 0.$$

Problem 10: Apply Boundedness. From the general solution in Problem 9, retain only the solutions bounded at $x = 0$.

Problem 11: Recognize A Half-Integer Order. Transform:

$$x^2 u'' + 2xu' + (x^2 - 2)u = 0$$

using $u = x^{-1/2}v$. Identify the Bessel order.

Problem 12: Extract A Boundary Condition. Suppose:

$$u(r) = AJ_2(\lambda r)$$

is nontrivial on $0 \leq r \leq 3$ and satisfies $u(3) = 0$. Write the allowed values of λ in terms of the positive zeros $j_{2,k}$ of J_2 .

Worked Answers: Problems 1 To 4**Answer 1.** For a positive integer n :

$$\Gamma(n) = (n-1)!.$$

Set $n = 8$:

$$\begin{aligned}\Gamma(8) &= (8-1)! \\ &= 7! \\ &= \boxed{5040}.\end{aligned}$$

Answer 2. Use the recurrence until the argument reaches $1/2$:

$$\begin{aligned}\Gamma\left(\frac{9}{2}\right) &= \frac{7}{2}\Gamma\left(\frac{7}{2}\right) \\ &= \frac{7}{2} \cdot \frac{5}{2}\Gamma\left(\frac{5}{2}\right) \\ &= \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2}\Gamma\left(\frac{3}{2}\right) \\ &= \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}\Gamma\left(\frac{1}{2}\right).\end{aligned}$$

Insert $\Gamma(1/2) = \sqrt{\pi}$:

$$\boxed{\Gamma\left(\frac{9}{2}\right) = \frac{105\sqrt{\pi}}{16}}.$$

Answer 3. Start with the recurrence at $s = -1/2$:

$$\Gamma\left(\frac{1}{2}\right) = -\frac{1}{2}\Gamma\left(-\frac{1}{2}\right).$$

Therefore:

$$\Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}.$$

Now use the recurrence at $s = -3/2$:

$$\Gamma\left(-\frac{1}{2}\right) = -\frac{3}{2}\Gamma\left(-\frac{3}{2}\right).$$

Divide by $-3/2$:

$$\begin{aligned}\Gamma\left(-\frac{3}{2}\right) &= \frac{-2\sqrt{\pi}}{-3/2} \\ &= (-2\sqrt{\pi})\left(-\frac{2}{3}\right) \\ &= \boxed{\frac{4\sqrt{\pi}}{3}}.\end{aligned}$$

Answer 4. Begin with:

$$\sum_{k=0}^{\infty} (k+1)a_k x^{k+3}.$$

Set:

$$j = k + 3.$$

From $j = k + 3$, solve for k and $k + 1$:

$$k = j - 3, \quad k + 1 = j - 2.$$

When $k = 0, j = 3$. Therefore:

$$\sum_{k=0}^{\infty} (k+1)a_k x^{k+3} = \sum_{j=3}^{\infty} (j-2)a_{j-3} x^j.$$

Worked Answers: Problems 5 To 8

Answer 5. For $\nu = 3$:

$$J_3(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\Gamma(m+4)} \left(\frac{x}{2}\right)^{2m+3}.$$

Calculate one term at a time:

$$\begin{aligned} m=0: & \quad \frac{1}{\Gamma(4)} \left(\frac{x}{2}\right)^3 = \frac{x^3}{48}, \\ m=1: & \quad -\frac{1}{\Gamma(5)} \left(\frac{x}{2}\right)^5 = -\frac{x^5}{768}, \\ m=2: & \quad \frac{1}{2!\Gamma(6)} \left(\frac{x}{2}\right)^7 = \frac{x^7}{30720}, \\ m=3: & \quad -\frac{1}{3!\Gamma(7)} \left(\frac{x}{2}\right)^9 = -\frac{x^9}{2211840}. \end{aligned}$$

Hence:

$$J_3(x) = \frac{x^3}{48} - \frac{x^5}{768} + \frac{x^7}{30720} - \frac{x^9}{2211840} + \dots$$

Answer 6. The $m = 0$ term of the defining series is:

$$J_\nu(x) \sim \frac{1}{\Gamma(\nu + 1)} \left(\frac{x}{2}\right)^\nu.$$

Set $\nu = 5/2$:

$$J_{5/2}(x) \sim \frac{1}{\Gamma(7/2)} \left(\frac{x}{2}\right)^{5/2}.$$

From Block 2:

$$\Gamma\left(\frac{7}{2}\right) = \frac{15\sqrt{\pi}}{8}.$$

Therefore:

$$\frac{1}{\Gamma(7/2)} = \frac{8}{15\sqrt{\pi}}.$$

The simplified leading behaviour is:

$$J_{5/2}(x) \sim \frac{8}{15\sqrt{\pi}} \left(\frac{x}{2}\right)^{5/2}.$$

Answer 7. The neighbour sum is:

$$J_{\nu-1} + J_{\nu+1} = \frac{2\nu}{x} J_\nu.$$

To involve J_1, J_2, J_3 , set $\nu = 2$:

$$J_1 + J_3 = \frac{4}{x} J_2.$$

From $J_1 + J_3 = (4/x)J_2$, subtract J_1 from both sides:

$$J_3(x) = \frac{4}{x}J_2(x) - J_1(x).$$

Answer 8. Use the neighbour-difference derivative identity:

$$J'_\nu = \frac{1}{2}(J_{\nu-1} - J_{\nu+1}).$$

Set $\nu = 2$:

$$J'_2(x) = \frac{1}{2}[J_1(x) - J_3(x)].$$

Worked Answers: Problems 9 To 12

Answer 9. The equation is:

$$x^2y'' + xy' + (25x^2 - 4)y = 0.$$

Compare it with:

$$x^2y'' + xy' + (\lambda^2x^2 - \nu^2)y = 0.$$

Match the parameters:

$$\lambda^2 = 25, \quad \nu^2 = 4.$$

Choose $\lambda = 5$ and $\nu = 2$. Set:

$$z = 5x, \quad y(x) = u(z).$$

Then:

$$y' = 5u_z, \quad y'' = 25u_{zz}, \quad x = \frac{z}{5}.$$

Substitute:

$$\left(\frac{z}{5}\right)^2 (25u_{zz}) + \left(\frac{z}{5}\right) (5u_z) + \left[25\left(\frac{z}{5}\right)^2 - 4\right] u = 0.$$

Simplify:

$$z^2 u_{zz} + zu_z + (z^2 - 4)u = 0.$$

This is Bessel's equation of order 2, so:

$$u = C_1 J_2(z) + C_2 Y_2(z).$$

Return to $z = 5x$:

$$\boxed{y(x) = C_1 J_2(5x) + C_2 Y_2(5x)}.$$

Answer 10. From Answer 9:

$$y(x) = C_1 J_2(5x) + C_2 Y_2(5x).$$

Because $J_2(5x)$ is bounded and $Y_2(5x)$ is singular at zero, boundedness requires:

$$C_2 = 0.$$

The bounded family is:

$$y(x) = C_1 J_2(5x).$$

Answer 11. The equation is:

$$x^2 u'' + 2xu' + (x^2 - 2)u = 0.$$

This is the spherical equation with:

$$\ell(\ell + 1) = 2.$$

The nonnegative integer solution is $\ell = 1$, since:

$$1(1 + 1) = 2.$$

Set $u = x^{-1/2}v$. The transformation derived in Block 7 gives:

$$x^2 v'' + xv' + \left[x^2 - \left(\ell + \frac{1}{2} \right)^2 \right] v = 0.$$

Substitute $\ell = 1$:

$$x^2 v'' + xv' + \left[x^2 - \left(\frac{3}{2} \right)^2 \right] v = 0.$$

The transformed equation has $(3/2)^2$ as its order term. Therefore the Bessel order is:

$$\nu = \frac{3}{2}.$$

Answer 12. The outer condition is:

$$u(3) = 0.$$

Substitute $u(r) = AJ_2(\lambda r)$:

$$AJ_2(3\lambda) = 0.$$

The solution is nontrivial, so $A \neq 0$. Divide by A :

$$J_2(3\lambda) = 0.$$

If $j_{2,k}$ is the k th positive zero of J_2 , then:

$$3\lambda = j_{2,k}.$$

Divide by 3:

$$\lambda_k = \frac{j_{2,k}}{3}, \quad k = 1, 2, 3, \dots$$

Chapter 30 Final Summary

The gamma function extends factorial structure:

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt, \quad \Gamma(s+1) = s\Gamma(s),$$

with:

$$\Gamma(n+1) = n!, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Bessel's equation of order ν is:

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

Its normalized first-kind Frobenius solution is:

$$J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m + \nu + 1)} \left(\frac{x}{2}\right)^{2m+\nu}.$$

The central identities are:

$$J_{\nu-1} + J_{\nu+1} = \frac{2\nu}{x} J_\nu,$$

$$J'_\nu = \frac{1}{2}(J_{\nu-1} - J_{\nu+1}).$$

For integer order, a full basis away from the origin is J_n, Y_n . Boundedness at the origin usually removes Y_n , while an outer boundary condition selects zeros of J_n or J'_n .

gamma notation packages the coefficients, Frobenius' method constructs the solution, recurrence identities make it usable, and boundary conditions select the physically admissible modes.